

Dinethra Rajapaksha

Electronic and Telecommunication Engineering Undergraduate, University of Moratuwa
[🌐 DinethraDivanjana2001](#) | [in Dinethra Rajapaksha](#) | [🌐 dinethrarajapaksha.vercel.app](#)
divanjanad@gmail.com | rajapakshaipdd.21@uom.lk | +94-704916740
Address: 113/5H, Siyambalape Waththa, Delgoda, Sri Lanka.

SUMMARY

Final-year undergraduate from the University of Moratuwa with a focus on software development, machine learning and Computer Vision supported by a strong background in embedded systems and IoT. Seeking opportunities to contribute to software and AI-driven engineering projects in innovative environments.

EXPERIENCE

Magicbits (PVT) Ltd

Dec 2024 – Jun 2025 — On-site

Software/IoT Engineering Intern

- Developed backend data-processing logic for real-time systems, handling ingestion, transformation, validation, and persistence of time-series data using MQTT, REST APIs, and rule-based workflows.
- Designed scalable data models and hierarchical aggregation pipelines to support multi-tenant dashboards, enabling reliable analytics and monitoring across 30+ production deployments.
- Implemented fault-tolerant logic for data quality assurance, including outlier detection, anomaly filtering, alarm decoding, and timestamp normalization to ensure correctness of downstream analytics.
- Built interactive dashboards and control workflows with clear separation of concerns between data processing, business logic, and presentation layers, supporting both real-time and historical analysis.

EDUCATION

University of Moratuwa, Moratuwa, Sri Lanka

July 2022 – Present

B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering GPA: 3.67 / 4.00

Dean's List: Semester 4, Semester 6, Semester 7

- *Relevant Coursework:* Data Structures and Algorithms, Applied Statistics, Software Design Competition, Robotics, Embedded Systems and Applications, Machine Learning, Deep Learning for Vision, Image Processing, Pattern Recognition, Neural Networks and Fuzzy Logic, Engineering Optimization, Autonomous Systems

Bandaranayake College, Gampaha, Sri Lanka.

January 2012 - October 2020

GCE Advanced Level - Physical Science Stream; Z-Score 2.5516. [🔗](#)

- *Combined Mathematics, Physics, Chemistry.*

GCE Ordinary Level ; 9As. [🔗](#)

PROJECTS

Quadruped Mobility System for Dynamic Sensing (Unitree Go2) — FYP [🔗](#)

(Ongoing)

- **Tools & Technologies:** Python, C++, CUDA, ROS 2 Humble, PyTorch, YOLOv11-OBB, YOLOv8, OpenCV, FastAPI, SQLite, Docker, TensorRT, NumPy, scikit-learn, EasyOCR, PaddleOCR, Vision-Language Models (VLMs), Jetson Orin Nano, Unitree Go2
- **Project Overview:** Designed a real-time visual inspection system for quadruped-mounted industrial monitoring, integrating GPU-accelerated oriented object detection, multi-stage analog gauge reading, and distributed edge-server workflows for autonomous inspection.

• My Contributions: Visual Inspection system

- Architected a distributed inspection pipeline with Jetson-based edge inference and an asynchronous FastAPI server for ROI-based job scheduling and result retrieval over Wi-Fi LAN.
- Developed a hybrid analog gauge reading system combining CNN and vision transformer-based keypoint regression, geometric ellipse fitting, segmentation-based needle detection, OCR with rotation normalization, and RANSAC-based angle-to-value regression.
- Integrated zero-shot VLM-based gauge reasoning with structured JSON parsing as a fallback pipeline, achieving 98.1% accuracy on unseen gauge types.

TransX – Transformer Maintenance Platform [↗](#)

November 2025

- Developed a full-stack AI-powered platform for transformer inspection, integrating React/TypeScript frontend, Spring Boot backend, and a YOLOv8-based Python ML microservice for real-time thermal anomaly detection and interactive annotation workflows.
- Implemented the complete database schema (MySQL, relational database), REST API endpoints, and frontend modules for inspection management, annotation canvas, and maintenance record generation.

Event-Driven Workflow Management Platform [↗](#)

(Ongoing)

- Designed and implemented a full-stack event-driven workflow platform using Apache Kafka for asynchronous task processing, featuring REST API with Express, Kafka-based event streaming, and React frontend with real-time status updates demonstrating microservices architecture and distributed systems patterns.
- Independently developed all system components: backend API with event publishing, Kafka consumer worker with step-by-step processing logic, React UI with auto-refresh capabilities, Docker-based infrastructure setup, and comprehensive technical documentation including architecture design, API specifications, and event schemas.

YOLO & Kalman filter based Target Tracking System [↗](#)

December 2025

- Designed and implemented an end to end video object tracking system combining YOLO-based detection with Kalman filter multi object tracking for robust, real-time tracking by detection, including occlusion handling and jitter reduction.
- Independently developed all modules: dataset preparation, model training, inference pipelines, Kalman tracking logic, visualization utilities, and automated result analysis with annotated videos and performance plots.

FasterViT-OCI: Boundary-Aware Hierarchical Vision Transformer Extension [↗](#)

December 2025

- Implemented and evaluated the FasterViT architecture, extending the original paper with OCI improvements to enhance hierarchical attention mechanisms; conducted three comprehensive experiments: inference throughput replication on ImageNet-1K, transfer learning on CIFAR-10, and object detection benchmarks with Detectron2 backbones on Pascal VOC.
- Designed and executed the full training and evaluation pipeline for FasterViT-0 on CIFAR-10, including data preprocessing, augmentation, checkpointing, and metric logging. Implemented the OCI extension in PyTorch, trained and evaluated both baseline and OCI models, and performed quantitative and qualitative comparisons.

Smart Rolling Door Device [↗](#)

August 2025

- Developed embedded firmware for ESP32-based controller, implementing MQTT communication, state management, fault handling, and real-time device telemetry for reliable remote operation.
- Designed and implemented a web-based control interface for door operation, live status visualization, and configuration management, focusing on usability and low-latency command execution.
- Integrated cloud-based messaging and event-driven workflows to enable secure remote access, automation triggers, and scalable device control.

Vision-Based Object Tracking and Control System (Robotic Arm) [↗](#)

July 2025

- Developed a real-time vision-based system that detects and tracks a selected target object and generates continuous control commands to align and follow the target using a robotic arm.
- Implemented the complete perception-to-control pipeline, including camera-based object detection and tracking, coordinate mapping from image space to actuator commands, and embedded control logic for smooth and stable object-following behavior.

CLRerNet Improving Confidence of Lane Detection with LaneloU [↗](#)

August 2024

- CLRerNet enhances lane detection confidence using the LaneloU architecture, improving accuracy in complex driving scenarios. The project focuses on advanced neural network techniques to refine lane recognition in real-time.
- Develop the CLRerNet algorithm using Python, leveraging the LaneloU architecture to improve lane detection confidence. The implementation will involve training and fine-tuning the neural network to enhance accuracy in detecting lanes under diverse driving conditions.

Hand Gesture Detection using machine learning and Image processing [↗](#)

June 2024

- This project harnesses the power of machine vision and deep learning to create a image recognition system.
- implemented a hand gesture detection system using CNNs, achieving high accuracy in gesture recognition. I optimized the model for real-time tracking and classification.

Ratatouille : Fast Maze Traversing Micro mouse [↗](#)

June 2024

- Developed an autonomous Micromouse robot capable of navigating unknown mazes using onboard sensors and real-time decision-making algorithms.
- Implemented maze-solving and motion-control algorithms in C++, including flood-fill-based path planning, wall detection and mapping, PID-based motor control, and efficient turn/acceleration profiles.

Path Planning in Robot Operating System ROS2 [↗](#)

August 2024

- This project implements the Dijkstra algorithm for path planning in ROS2, enabling efficient autonomous robot navigation by finding the shortest path while avoiding obstacles.
- Implement the Dijkstra algorithm in ROS2 for autonomous path planning by integrating the navigation stack with real-time obstacle avoidance.

Industrial Smart Soldering Station [↗](#)

March 2024

- The Smart Soldering Station is a high-performance soldering solution with a soldering iron and hot air gun, featuring advanced PID temperature control and multiple modes for precise, efficient, and reliable soldering.
- Firmware development, Research on the circuit, designed AC and DC circuit schematics and PCBs, developed PID control algorithm, selected components, and assisted in enclosure design soldering and testing.

TechbBot : STM32/Vision-Based Robot [↗](#)

March 2023

- An innovative robot featuring a custom-designed 4-layer PCB with an STM32F405VGT6 ARM-M4 microcontroller and a Raspberry Pi 4b for advanced machine vision tasks.
- Designed the shooter mechanism, worked with Cube IDE for development, tested sensors, assembled mechanical and electronic components, and was involved in algorithm design using C++.

IoT Based Medicine Storage [↗](#)

May 2023

- The Smart MediBox assists users in managing their medication schedules with timely audio and visual reminders, environmental tracking, and remote management via a web-based dashboard.
- Independently designed and developed the circuit, PCB, and dashboard designed using Node-RED, and programmed(c++) ESP32 microcontroller using the Wokwi simulator.

Turbi-Detector-Iot Base Solution for Contaminated Water [↗](#)

July 2023

- The roject aims to develop a smart device using an infrared sensor and control system to detect mud in water pipelines. It ensures clean water supply by diverting contaminated water and notifying users via a mobile app.
- Firmware development, Designed the circuitry, schematics, and PCB layout for reliable operation. Created graphic designs for marketing, algorithm-designed , system integration.

Nanonaut : Tasks Completing Line Following robot [↗](#)

December 2024

- The Nanonaut Robotics Project features a robot capable of accurate line following, intelligent wall avoidance, ramp navigation, arm operation, sound responsiveness, color detection, maze traversal, and obstacle avoidance.
- Designed the robot's electronics and mechanical assembly, selected sensors and components, implemented sensor algorithms, and contributed to 3D design for optimal functionality and performance.

UART Transceiver Implementation in FPGA [↗](#)

May 2023

- Implemented a UART transceiver using FPGA, enabling reliable serial communication for data transmission and reception.
- Testing the transceiver for accurate data handling and simulate the idea.

AWARDS AND SCHOLARSHIPS

Finalist — Sri Lanka Robotic Challenge 2024 [↗](#)

Competed in the prestigious Sri Lanka Robotic Challenge 2024, demonstrating advanced robotic design and programming skills. Achieved finalist status by successfully navigating through complex challenges and showcasing innovative solutions in the university category.

Mahapola Higher Education (Merit) Scholarship.

For outstanding performance in the GCE A/L Examination. Ranked #124 out of 34389 students.

TECHNICAL SKILLS AND INTERESTS

Languages: English (Professional), Sinhala (Native)

Programming: Python, C/C++, MATLAB

Developer Tools: Altium Designer, SolidWorks, LTspice, Multisim, Quartus Prime, Arduino, Atmel Studio, PIS-tudio, Webots, Version Control (Git and GitHub), SLAM toolbox, TensorFlow

Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn

Cloud / Databases: MySQL, Firebase

Soft Skills: Problem Solving, Decision Making, Project Management, Team Leadership, Marketing

Coursework: Machine Vision, Deep Learning, Applied Machine Learning, Engineering Project Management

Areas of Interest: Software development, Deep Learning for Computer Vision, Machine Learning, Embedded Systems, Electronic Product Design

Sports & Other: Badminton, Playing the Guitar

CERTIFICATIONS

Deep Learning & Machine Learning: Convolutional Neural Networks (DeepLearning.AI, 2025), Structuring Machine Learning Projects (DeepLearning.AI, 2025), Supervised Machine Learning: Regression and Classification (DeepLearning.AI, Stanford University, 2024)

Programming & Problem Solving: Problem Solving – Basic (HackerRank, 2024), Problem Solving – Intermediate (HackerRank, 2024), Python – Basic (HackerRank, 2024)

VOLUNTEER EXPERIENCE

Career Fair 2023 - University of Moratuwa

- Company Coordinator : ADL(Axiata Digital Lab)

Sri Lanka Robotic Challenge 2023 - University of Moratuwa

- Coordinator : Video Streaming

Abhina 2023 - University of Moratuwa

- Coordinator : Video Streaming

Exmo exhibition - University of Moratuwa

- Coordinator

Rotaract Club of University of Moratuwa.

- *General Member* *June 2022 - April 2023*
- *Engaged as an organizing committee member for various projects(Professional Development projects, Membership Development projects, Community Service projects)*

Science Society - Bandaranayake College, Gampaha

- Coordinator : Organized and led various science-related activities and events, including the Xban Exhibition.

Scout Association - Bandaranayake College,Gampaha

Buddhist Society - Bandaranayake College,Gampaha

Xban Exhibition 2018 and 2014 - Bandaranayake College, Gampaha

- Coordinator (2014): Led the coordination of an astronomy-related project, organizing team activities, overseeing project development, and ensuring the successful presentation of the project.
- Coordinator (2018): Managed and coordinated a project related to pyramid technology, facilitating team collaboration, guiding project execution.

REFERENCES

Dr. Ranga Rodrigo

Senior Lecturer
Department of Electronic and Telecommunication
Engineering,
University of Moratuwa, Sri Lanka
Email : ranga@uom.lk

Peshala Jayasekara,PhD

Senior Lecturer
Department of Electronic and Telecommunication
Engineering,
University of Moratuwa, Sri Lanka
Email : peshala@uom.lk